SALTDRAG · PRODUCT REQUIREMENTS AND SPECIFICATIONS

SaltDrag product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-24	Amish Chadha	Kit 1.8 upgrade; SDG-REQ-001 becomes SDG-PRS-001; R1 to R12 migrated to PR-001 to PR-012 with tolerance, units and the user requirement each traces to; PR-013 to PR-018 added; status from SDG-CAL-001 v0.3 after the STEM check fixes (SDG-DEC-001 D17 to D24); the assumption that several families share one set is withdrawn
0.3	2026-10-03	Amish Chadha	TRL 3 status from SDG-CAL-001 against the constructable design (SDG-DDR-002); R5 not verified
0.2	2026-10-03	Amish Chadha	TRL 2; every target made measurable; R10 to R12 added for rope, holding and rope capacity (SDG-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold (as SDG-REQ-001)

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What the design must do, each requirement traced to a user need in SDG-URS-001 (`03a-user-requirements.md`); the links and the evidence are in the traceability matrix SDG-RTM-001 (`03b-traceability.md`). Seventeen of the eighteen requirements are met on paper or by design; PR-005, the gathering rate against the dantala, cannot be shown until the dantala baseline is measured on a partner's pan. Status is from the calculation note SDG-CAL-001 v0.3 (estimates). Physical verification is TRL 4 and TRL 5 work, and the targets remain proposals for review with a co-design partner. Method: STANDARDS section 22. This document replaces SDG-REQ-001; the old R numbers are in the "Was" column.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Usability	Work from the bund	A full harvest pass, including set-up and moving to the next strip, shall need no worker to enter the pan	0	max	workers in the pan	Analysis (layout), TRL 3; observed field trial on a partner pan, TRL 5	Met by design	R1
PR-002	UR-002	Usability	Crank effort	The force at each of two crank handles at the design rake load shall not exceed the target	120	max	N	Analysis (CAL-001 [B3]), TRL 3; spring scale on the handle, TRL 4	Met on paper	R2
PR-003	UR-003, UR-009	Performance	Pull capacity and margin	Two people at 120 N shall give at least the target rope pull, and every load-bearing part shall keep a margin of 2 on yield or rated strength at twice the target	3.0	min	kN	Analysis (CAL-001 [B4], [B5], Section C), TRL 3; proof load with CalRig, TRL 4	Met on paper	R3
PR-004	UR-004	Physical architecture	Protect the pan floor	The blade edge shall ride 5 to 25 mm above the floor, no steel shall touch the pan, and skid pressure shall not exceed 15 kPa	15	max	kPa	Analysis (CAL-001 [D1], [D2]), TRL 3; inspection after 20 passes, TRL 5	Met by design	R4
PR-005	UR-003	Performance				min				R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
			Gathering rate	The set shall gather at least as many kilograms per worker-hour as the dantala method on the same pan	dantala rate		kg per worker-hour	Side-by-side timed harvest trial, TRL 5	Not met (not verified; no baseline)	
PR-006	UR-005	Packaging and transport	Part mass	No single part shall exceed the target, and every part of the set shall be carried by up to four people	25	max	kg	Analysis (model masses, CAL-001 [E1]), TRL 3; weigh parts, TRL 4	Met on paper	R6
PR-007	UR-006	Materials	Corrosion resistance	The set shall keep working after one season in brine and salt with a daily freshwater rinse	1	min	season	Analysis (materials, SDG-STM-001 S18 to S20), TRL 3; season trial with inspection log, TRL 5	Met by design (316 stainless in brine to confirm, S19)	R7
PR-008	UR-007	Usability	Quick set-up	Capstans anchored and rake rigged by two crews working at once within the target	20	max	min	Analysis (CAL-001 [D10]), TRL 3; timed trials, TRL 4	Met on paper, small margin	R8
PR-009	UR-008	Physical architecture	Value engineering	The parts cost of the full set shall be reported against the value-engineering target	4,000	max	USD	Analysis (costed BOM, CAL-001 [E2]), TRL 3	Met on paper (USD 2,897)	R9
PR-010	UR-009	Safety		The rope's minimum	5	min		Analysis (CAL-001		R10

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
			Rope safety	breaking strength after splices shall be at least the target multiple of the rated 3 kN pull			times the rated pull	[C8]), TRL 3; rope certificate and proof load, TRL 4	Met on paper	
PR-011	UR-009	Safety	Holding	The pawl shall hold 6 kN; the brake shall hold 3 kN with a hand force of 150 N or less; each ground anchor shall hold at least 1.5 times its share of the rated pull	150	max	N on the brake lever	Analysis (CAL-001 [C5b], [C7], [C10b]), TRL 3; proof load and anchor pull test on site, TRL 4	Met on paper; soil to confirm	R11
PR-012	UR-012	Performance	Pan coverage	Each drum shall store at least the target length of rope, for pans up to 30 m between bund crests	45	min	m	Analysis (CAL-001 [B8]), TRL 3; wind on and measure, TRL 4	Met on paper	R12
PR-013	UR-010	Physical architecture	Hand power only	The set shall have no engine, motor or stored-energy drive	0	max	W of non-human drive power	Inspection of the design (CAL-001 Section G), TRL 3	Met by design	
PR-014	UR-013	Physical architecture	Nothing left in the sanctuary	Every part on site shall be removed with hand tools when the set moves, leaving no structure	0	max	fixed structures left	Inspection of the design (CAL-001 Section G; SDG-BLD-001), TRL 3; demonstration	Met by design	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
								at the trial pan, TRL 5		
PR-015	UR-011	Materials	Local stock	Every made part shall be cut, drilled and welded from plate, hollow section, bar and sheet sold in Gujarat; every bought part shall be a catalogue item from an Indian stockist	100	min	% of BOM lines	Inspection of the BOM (CAL-001 Section G), TRL 3; build, TRL 4	Met by design	
PR-016	UR-014	Usability	Crew size	The set shall be worked by a crew of no more than the target	4	max	people	Analysis (CAL-001 [D9], Section G), TRL 3; trials, TRL 5	Met by design	
PR-017	UR-015	Usability	Daily care	A daily freshwater rinse and weekly grease shall keep the set working, with spares for the pawl, spring, lining and shackles in the kit	15	max	min of care a day	Inspection of the BOM and build plan (CAL-001 Section G), TRL 3; season trial, TRL 5	Met by design	
PR-018	UR-002, UR-009	Safety	Cranking power	Each of two crankers shall work at no more than the target rate on the loaded pull	75	max	W	Analysis (CAL-001 [B7b]; SDG-STM-001 S15), TRL 3; heart-rate and timing trial with the partner, TRL 5	Met on paper (71 W)	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- The drag of a rake on a crystallised bed is about 0.9 kN per metre of blade with a full heap (SDG-CAL-001, Section A); this is the main uncertainty and is measured with the partner before the first build is loaded (SDG-STM-001 S1).
- Hand capstans are not treated as machines under the Agariya access conditions, which bar heavy machinery; written clarification is sought through the partner before any trial (SDG-REG-001).
- One set serves one family pan. Version 0.3 assumed several families could share a set; at about 195 t a harvest one set gathers about a third of what one pan yields in a year (SDG-CAL-001 [D11], [D12]; SDG-DEC-001 D20), so that assumption is withdrawn.
- Salt gathered into the windrow, about 3.5 m from the bund toe, is loaded with existing methods (out of scope).
- The pans have dry, firm ground behind the bunds on the two sides where the capstans stand.

> **Safety:** PR-003, PR-010, PR-011 and PR-018 are safety requirements. A load-bearing part that misses its margin, a rope without a certificate, or an anchor that fails its proof load on site stops the work until it is replaced (SDG-BLD-001, Safety stops). Heat is the largest everyday hazard in the March and April harvest; cranking is planned for the cooler hours (SDG-STM-001 S16).